

Report 05

Risk Engine: The Two-Stage Constraint Hierarchy

Project:	JaJeev — Agentic Marine Intelligence System
Phase:	Phase 0, Week 3 (Sep 8–14, 2026) — with critical bug-fix on Aug 26
Focus:	Risk scoring model, hard constraint vetoes, vessel-aware safety

1. The Bug That Changed Everything

On **August 26, 2026**, a review of the risk engine revealed a real, potentially lethal design defect — not a hypothetical one.

1.1 The Original Design (Flat Additive Score)

`risk_score = w1×wave + w2×wind + w3×cyclone + ... + w_mpa×20`

Under this design, an MPA violation contributed only **+20 points**. This meant a spot *inside a legally protected area* could still be returned as **"LOW risk" (score < 25)** if every other factor happened to be favourable.

A legal prohibition was competing on a numeric scale with wave height, and losing.

1.2 The Fix: Two-Stage Constraint Hierarchy

The risk engine was redesigned into two mandatory stages:

Stage 1 — Hard Constraint Vetoes

Runs **first**, independently of any scoring. A candidate that trips any of these is immediately returned as `risk_level="REJECTED"` — it is **never scored**:

Hard Constraint	When It Fires
Marine Protected Area	Point is inside an MPA boundary
Certain-capsize wave height	Wave height exceeds vessel's absolute maximum
Grounding risk	Water depth < vessel draft + safety margin
On land	GEBCO reports depth ≤ 0 (terrestrial)
Out of range	Distance from nearest port exceeds operational range

Stage 2 — Soft Factor Scoring (Survivors Only)

Only runs on candidates that pass Stage 1. 0–100 additive score, every point individually explainable:

`risk_score = (`

```

wave_factor(wave_height, vessel.max_safe_wave_m) +
wind_factor(wind_speed, vessel.wind_threshold_ms) +
cyclone_factor(cyclone_distance_km) +
current_factor(current_speed) +
precipitation_factor(rainfall_mm) +
visibility_factor(visibility_m) +
pressure_trend_factor(pressure_trend)
)

```

Risk Tiers

Tier	Score	Meaning
LOW	0–24	Safe to proceed
MODERATE	25–49	Proceed with caution
HIGH	50–74	Strongly advise against departure
EXTREME	75–100	Do not proceed
REJECTED	(none)	Stage 1 veto. Not permitted or not survivable.
INSUFFICIENT_DATA	(none)	Too many inputs missing. Refuses to guess.

The last two are **not points on the scale** — they are separate outcomes, and must be presented as such.

2. Vessel-Aware Risk Scoring

2.1 The Problem

The original system assumed one implicit vessel — an 8m fishing boat — with thresholds baked into every calculation. But a calm day for a 200m bulk carrier is a capsized event for an 8m boat.

2.2 The Solution: 11 Named Vessel Classes

Three distinct user groups, each with physically different risk profiles:

Group	Vessels	Unique Risk
Fishermen	fishing_small (8m), fishing_mechanized (18m)	Wave height, operational range
Sailors	sailing_yacht (12m), dhow (20m)	Becalmed risk, point-of-sail
Traders	coastal_trader (40m) – container_ship (200m)	Squat, under-keel clearance

2.3 Sail-Specific Risk Factors

For sailing vessels:

- **Becalmed:** Wind below minimum working threshold = no drive, no steerage
- **Point of sail:** Course falls inside the vessel's no-go angle
- Motor vessels **never** trigger either — sail fields are None

2.4 Commercial Vessel Depth Rules

- **Squat:** Barrass approximation: $\text{squat}_m = C_b \times V^2 / 100$
- **Under-keel clearance:** 10% of static draft (PIANC-style)
- **Tide correction:** EOT20 tide height subtracted from charted depth

3. Validation Agent

A safety net that checks the Risk Agent's output for internal contradictions:

Check	Catches
vetoed=True + risk_level != "REJECTED"	Logic error in constraint pipeline
risk_score > 100 or < 0	Out-of-bounds arithmetic
All factor values are positive	Negative contributions indicate bug
sources_used is not empty	Silent data drop

4. Testing: 93 Pytest Tests

The risk hierarchy is regression-tested in tests/test_risk_agent.py:

- Same sea conditions, different vessels → different risk levels
- MPA violation → REJECTED regardless of other factors
- Wave height above capsize threshold → REJECTED
- Land point → REJECTED
- All factors calm → LOW
- Multiple hazards stacked → correct additive scoring

5. Evidence Agent

Every recommendation is accompanied by a deterministic EvidenceReceipt:

```
class EvidenceReceipt:
    sources_used: list[str]
    data_freshness: dict[str, str]
    factor_breakdown: dict[str, int]
    explanation: list[str]
    data_gaps: list[str]
    rejected_alternatives: list[str]
    confidence_note: str
```

The LLM only phrases this receipt — it never computes its contents. Both React and Android render the same receipt object, so explanation cannot drift from computation.

6. Impact

This two-stage design is listed as **Innovation #11** in the PRD:

"A legal or survivability constraint cannot be outvoted by a good score. Most AI risk scoring collapses legality and preference onto one number; this system refuses to score a candidate it has already ruled out, and says so."